

STATIC WEBSITE HOSTING USING AMAZON S3 WITH GITHUB

Prepared in the partial fulfillment of the Summer Internship Program on AWS

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Under the guidance of

Mrs. Sumana Bethala, APSSDC

Mr. Juttuka Lovababu, APSSDC

Submitted by

RAVVA SWAPNA, 24KT5A0519, PSCMR College of Engineering and Technology,
Vijayawada.

SHAIK ASHIK, 23KT1A05H8, PSCMR College of Engineering and Technology,
Vijayawada.

PUNNA POORNA CHANDU, 23KT1A05, PSCMR College of Engineering and Technology,
Vijayawada.

THAMMA GAYATHRI, 24KT5A0506, PSCMR College of Engineering and Technology,
Vijayawada.

IPPILI BHUMIKA, 24KT5A0503, PSCMR College of Engineering and Technology,
Vijayawada.

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Thank you.

Sincerely,

RAVVA SWAPNA, 24KT5A0519, ravvaswapna19@gmail.com.

SHAIK ASHIK, 23KT1A05H8, ashikshaik101@gmail.com.

PUNNA POORNA CHANDU, 23KT1A05H3, poornachandu7005@gmail.com.

THAMMA GAYATHRI, 24KT5A0506, thamma1580@gmail.com

IPPILI BHUMIKA, 24KT5A0503, ippilibhumika2@gmail.com.

➤ ABSTRACT

Cloud computing has become one of the most important technologies for hosting websites and applications because it provides scalability, reliability, security, and cost-effectiveness. Traditional web hosting requires dedicated servers, continuous maintenance, and higher operational costs. Cloud-based hosting overcomes these challenges by offering flexible, highly available, and low-cost solutions for deploying websites and applications.

This project, "**Static Website Hosting Using Amazon S3 with GitHub**," demonstrates how a static website can be developed and hosted using **Amazon Simple Storage Service (Amazon S3)**. The primary objective of this project is to provide a secure, reliable, and scalable platform for hosting static web pages while utilizing GitHub as a repository for storing and managing the website source code. The website is developed using standard web technologies such as **HTML, CSS, and JavaScript**, making it lightweight, responsive, and easy to maintain.

Amazon S3 is a highly durable and scalable object storage service provided by Amazon Web Services (AWS). It enables users to host static websites without the need for a traditional web server. In this project, an S3 bucket is created and configured for static website hosting. Website files, including HTML pages, CSS stylesheets, JavaScript files, and images, are uploaded to the bucket. Appropriate bucket policies and permissions are configured to allow public access to the hosted website while maintaining secure management of AWS resources.

GitHub is used as a version control platform to store and manage the project source code. It enables developers to organize project files, track changes, maintain different versions of the website, and collaborate effectively during development. By using GitHub, the project ensures proper source code management and simplifies future updates to the website.

The implementation process includes creating an AWS account, configuring an Amazon S3 bucket, enabling static website hosting, uploading website files, configuring bucket permissions, creating a GitHub repository, and maintaining the project source code. After successful configuration, the website is accessed through the Amazon S3 Static Website Endpoint and tested to verify proper functionality and accessibility.

The project demonstrates how Amazon S3 provides a cost-effective, reliable, and scalable solution for hosting static websites. It offers high availability, excellent durability, and easy maintenance without requiring server administration. This project also provides practical knowledge of AWS cloud services, static website hosting, and GitHub version control. The developed solution is suitable for hosting personal portfolios, educational websites, business landing pages, and other static web applications. Overall, this project successfully demonstrates the practical implementation of cloud computing technologies for efficient static website hosting using Amazon S3 and GitHub.

The project also emphasizes the importance of cloud-based storage and website management in modern web development. By utilizing Amazon S3, users can host websites with high durability, availability, and performance while eliminating the need for physical servers or complex infrastructure. AWS automatically manages the storage resources, ensuring that the website remains accessible even during high traffic conditions. The simple deployment process and minimal maintenance requirements make Amazon S3 an ideal solution for students, developers, startups, and organizations looking for an affordable and reliable web hosting platform.

This project provides valuable hands-on experience in Amazon Web Services (AWS), cloud storage concepts, and GitHub version control. It helps learners understand the practical implementation of static website hosting and the benefits of cloud computing in real-world applications. The knowledge gained through this project can be extended to more advanced cloud technologies such as content delivery networks, custom domain integration, HTTPS security, and automated deployment. Overall, the project establishes a strong foundation in cloud computing and web hosting while preparing students for future development of scalable and secure cloud-based applications.

TABLE OF CONTENTS

S.No	Content	Pg.No
1.	INTRODUCTION	5
2.	METHODOLOGY.....	6
3.	TECHNOLOGY STACK.....	8
4.	SYSTEM DESIGN / ARCHITECTURE	13
5.	IMPLEMENTATION.....	13
6.	RESULTS.....	16
7.	CONCLUSION.....	20

1. INTRODUCTION

Cloud computing has become one of the most important technologies in today's digital world, enabling organizations and individuals to store, manage, and access data and applications over the internet without maintaining physical infrastructure. It provides flexible, scalable, reliable, and cost-effective solutions for hosting websites, applications, and digital content.

Amazon Web Services (AWS) is one of the world's leading cloud computing platforms, offering a comprehensive range of cloud services, including computing, storage, networking, security, and database management. AWS enables developers and organizations to build, deploy, and manage applications efficiently without investing in expensive hardware or server infrastructure. Among its many cloud services, **Amazon Simple Storage Service (Amazon S3)** is one of the most widely used storage services for hosting static websites because of its durability, scalability, and ease of use.

A static website consists of fixed web pages created using technologies such as **HTML, CSS, JavaScript, images, videos, and other static resources**. Unlike dynamic websites, static websites do not require server-side programming languages or databases to generate content. As a result, they are faster, more secure, easier to maintain, and less expensive to host. Static websites are commonly used for personal portfolios, educational websites, business landing pages, company profiles, product showcases, event pages, and documentation websites.

Amazon S3 provides an efficient platform for hosting static websites by allowing users to store website files in an S3 bucket and make them publicly accessible through the Static Website Hosting feature. Once the website files are uploaded and the necessary permissions are configured, Amazon S3 generates a website endpoint that enables users to access the hosted website through any web browser. This eliminates the need for traditional web servers while ensuring high availability, fault tolerance, and excellent performance.

This project, "**Static Website Hosting Using Amazon S3 with GitHub**," demonstrates how a static website can be designed, managed, and hosted using AWS cloud services. The website is developed using HTML, CSS, and JavaScript and is uploaded to an Amazon S3 bucket

configured for static website hosting. Appropriate bucket policies are applied to allow users to access the website securely through the internet.

GitHub is used in this project as a version control and source code management platform. It provides a secure repository for storing the website's source code, tracking modifications, maintaining different versions of the project, and enabling effective collaboration among developers. Using GitHub ensures that the website files are organized, backed up, and easily manageable throughout the development process.

The implementation of this project involves creating an AWS account, configuring an Amazon S3 bucket, enabling static website hosting, uploading website files, configuring bucket permissions, creating a GitHub repository, and managing the website source code. After deployment, the website is accessed through the Amazon S3 Website Endpoint and tested to ensure that all pages, images, stylesheets, and scripts function correctly.

One of the major advantages of using Amazon S3 is its ability to provide highly durable and scalable storage without requiring server maintenance. AWS automatically manages the underlying infrastructure, allowing users to focus on website development rather than hardware management. The service also follows a pay-as-you-use pricing model, making it an affordable hosting solution for students, startups, and small businesses.

This project provides practical knowledge of cloud computing concepts, AWS services, website deployment, and version control using GitHub. It helps learners understand how cloud technologies simplify website hosting while improving accessibility, reliability, scalability, and security. The knowledge gained through this project also serves as a foundation for learning advanced AWS services such as Amazon CloudFront, Route 53, AWS Certificate Manager, and serverless web application development.

2. METHODOLOGY

The project "**Static Website Hosting Using Amazon S3 with GitHub**" was developed using a systematic methodology to ensure successful implementation and deployment. Initially, the project requirements were analyzed, and the necessary tools and technologies were identified. The website was designed and developed using **HTML, CSS, and JavaScript** to create a responsive and user-friendly interface. An **AWS account** was created, and an **Amazon S3 bucket** was configured for static website hosting. The **Static Website Hosting** feature was enabled, and the **index.html** file was set as the default homepage. Appropriate bucket policies and permissions were configured to allow public access to the website. All website files, including HTML, CSS, JavaScript, and images, were uploaded to the S3 bucket. **GitHub** was used to store, organize, and manage the project source code through version control. The website was then accessed using the **Amazon S3 Website Endpoint**. Various tests were conducted to verify that all web pages, images, hyperlinks, and styles were functioning correctly. The website was also tested on different web browsers and devices to ensure compatibility and responsiveness. Any issues identified during testing were resolved successfully. Finally, the website was deployed and made publicly accessible. The project demonstrated that Amazon S3 provides a reliable, scalable, secure, and cost-effective solution for hosting static websites while GitHub simplifies source code management.

2.1 Requirement Gathering

The requirement gathering phase focused on identifying the resources needed for developing and hosting the static website. The website was planned to be developed using **HTML, CSS, and JavaScript**. **Amazon Web Services (AWS)** was selected as the cloud platform for hosting the website. **Amazon S3** was chosen because it provides reliable, scalable, and cost-effective static website hosting. An AWS account was created to access and configure cloud services. **GitHub** was selected to store and manage the project source code using version control. A code editor such as **Visual Studio Code** and a stable internet connection were also required. AWS documentation and project requirements were studied before implementation. This phase

ensured that all necessary software, tools, and cloud resources were available before development began.

2.2 Design

The system was designed to provide a simple and efficient solution for hosting a static website using **Amazon S3**. The website interface was developed using **HTML, CSS, and JavaScript** to create an attractive and responsive design. An **Amazon S3 bucket** was created to store the website files. The **Static Website Hosting** feature was enabled to make the website accessible over the internet. Bucket policies were configured to allow public access to the website. **GitHub** was used to manage and organize the source code. The design focused on simplicity, reliability, scalability, and easy maintenance. The website was planned to be compatible with different web browsers and devices. This design ensured efficient cloud-based website hosting.

2.3 Implementation

The implementation phase involved creating an **Amazon S3 bucket** and enabling **Static Website Hosting**. The website was developed using **HTML, CSS, and JavaScript**. All website files, including images and stylesheets, were uploaded to the S3 bucket. Appropriate bucket permissions were configured to allow public access. A **GitHub repository** was created to store and manage the project source code. The source code was uploaded to GitHub using Git. The website was accessed through the **Amazon S3 Website Endpoint**. All webpages and resources were verified after deployment. The implementation was completed successfully following AWS best practices.

2.4 Storage Management

Amazon Simple Storage Service (Amazon S3) was used as the primary storage solution for the project. All website files, including **HTML, CSS, JavaScript, images, and other resources**, were stored in the S3 bucket. The bucket was configured for **Static Website Hosting**. Appropriate bucket policies were applied to allow users to access the website. Amazon S3 provides **high durability, scalability, and availability** for storing website files. It automatically manages storage without requiring physical servers. The hosted files remain

secure and easily accessible. Storage management through Amazon S3 reduces maintenance effort and operational costs. This approach provides a reliable cloud-based hosting solution.

2.5 Testing

The website was tested after deployment to ensure proper functionality. All webpages, images, CSS files, and JavaScript files were verified. The website was accessed using the **Amazon S3 Website Endpoint**. Navigation menus and hyperlinks were tested for correct operation. The website was checked on different web browsers such as Chrome and Edge. It was also tested on different screen sizes to ensure responsiveness. Any identified issues were corrected immediately. The loading speed and accessibility of the website were verified. Successful testing confirmed that the website functioned correctly and was ready for public access.

2.6 Deployment

The deployment phase involved hosting the completed website on **Amazon S3**. A unique S3 bucket was created and configured for **Static Website Hosting**. All website files were uploaded successfully to the bucket. The **index.html** file was configured as the default homepage. Bucket policies were updated to allow public access to the website. The website was published using the **Amazon S3 Website Endpoint**. The deployed website was verified to ensure proper accessibility. All pages, images, and links loaded successfully. The deployment demonstrated that Amazon S3 provides a simple, scalable, and cost-effective solution for hosting static websites.

➤ 4. TECHNOLOGY STACK

The project "**Static Website Hosting Using Amazon S3 with GitHub**" was developed using modern web technologies and AWS cloud services. The following technologies and tools were used during the implementation of the project.

Cloud Platform

- **Amazon Web Services (AWS):** AWS provides a secure, reliable, and scalable cloud platform for hosting the static website.

AWS Services

- **Amazon S3 (Simple Storage Service):** Used to store and host the static website files such as HTML, CSS, JavaScript, images, and other resources.
- **AWS Identity and Access Management (IAM):** Used to manage user permissions and provide secure access to AWS resources.

Front-End Technologies

- **HTML:** Used to create the structure and content of the web pages.
- **CSS:** Used to design the layout, colors, fonts, and responsive appearance of the website.
- **JavaScript:** Used to add interactivity, dynamic behavior, and improve the user experience.

Version Control

- **Git:** Used to track changes in the project and manage source code efficiently.
- **GitHub:** Used as a remote repository to store, organize, and maintain the website source code.

Development Tools

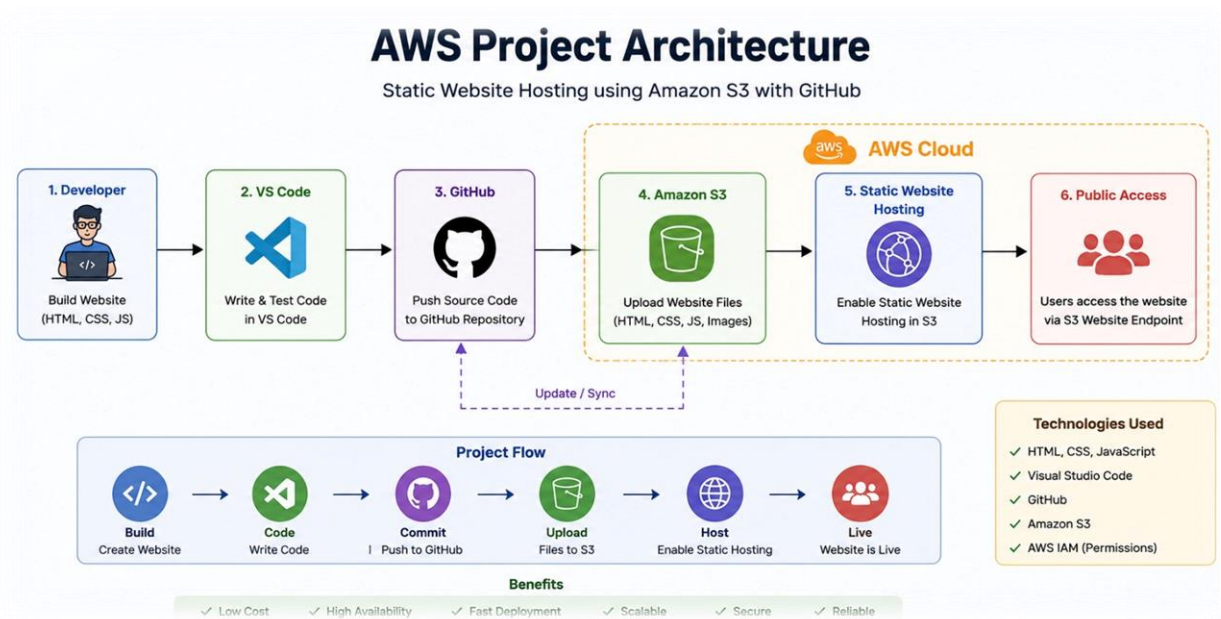
- **Visual Studio Code (VS Code):** Used as the source code editor for developing and modifying the website.
- **Web Browser (Google Chrome/Microsoft Edge):** Used to test and verify the functionality of the hosted website.

Website Hosting

- **Amazon S3 Static Website Hosting:** Used to publish the website and make it accessible through the Amazon S3 Website Endpoint.

➤ 4. SYSTEM DESIGN / ARCHITECTURE

The **Static Website Hosting Using Amazon S3 with GitHub** project follows a simple cloud-based architecture that separates website development, source code management, storage, and user access. The website is developed using **HTML, CSS, and JavaScript**, and the source code is stored and managed using **GitHub**. The completed website files are uploaded to an **Amazon S3 bucket**, where the **Static Website Hosting** feature is enabled. Amazon S3 hosts the website and provides a public website endpoint, allowing users to access the website through any web browser. This architecture is secure, scalable, reliable, and cost-effective, making it suitable for hosting static websites without requiring a traditional web server..



The architecture of the "**Static Website Hosting Using Amazon S3 with GitHub**" project illustrates the complete workflow from website development to public access. It demonstrates how the website is created, managed, stored, and hosted using AWS cloud services and GitHub. The architecture consists of six major components.

1. Developer

The project begins with the developer, who is responsible for designing and developing the static website. The website is created using **HTML, CSS, and JavaScript**, which define the structure, styling, and functionality of the web pages. The developer ensures that the website is responsive, user-friendly, and ready for deployment.

2. Visual Studio Code (VS Code)

Visual Studio Code is used as the source code editor for writing, editing, and testing the website files. It provides features such as syntax highlighting, debugging, and extensions that help in efficient web development. The developer uses VS Code to create HTML pages, CSS stylesheets, JavaScript files, and other website resources before uploading them.

3. GitHub

After the website is developed and tested, the project files are uploaded to a **GitHub repository**. GitHub acts as a version control system that securely stores the source code, tracks changes, and maintains different versions of the project. It also makes future modifications and collaboration easier by keeping all project files organized in a central repository.

4. Amazon S3

The website files stored in GitHub are manually uploaded to an **Amazon S3 bucket**. Amazon S3 serves as the cloud storage service that stores all static website resources, including HTML, CSS, JavaScript, images, and other files. The bucket is configured to host the website by enabling the **Static Website Hosting** feature.

5. Static Website Hosting

Once the website files are uploaded, **Static Website Hosting** is enabled in the Amazon S3 bucket. The **index.html** file is configured as the default home page, and the required bucket policies are applied to allow public access. Amazon S3 generates a **Website Endpoint URL**, which acts as the public address of the website.

6. Public Access

Users can access the website through the **Amazon S3 Website Endpoint** using any web browser. Since the website is hosted on AWS cloud infrastructure, it provides high availability, scalability, and reliable performance. Users can view all webpages, images, styles, and other static content without requiring a traditional web server.

Project Flow

The overall workflow of the project is as follows:

Build Website → Write Code in VS Code → Upload Source Code to GitHub → Upload Website Files to Amazon S3 → Enable Static Website Hosting → Website Becomes Live → Users Access the Website

Benefits of the Architecture

- **Low Cost:** Amazon S3 provides an affordable hosting solution with a pay-as-you-use pricing model.
- **High Availability:** The website remains accessible through AWS cloud infrastructure.
- **Easy Deployment:** Website files can be uploaded and updated easily in the S3 bucket.
- **Scalability:** Amazon S3 automatically handles increasing storage and user traffic.
- **Security:** AWS IAM and bucket policies help protect cloud resources.
- **Reliability:** Amazon S3 offers high durability and ensures website files are safely stored.

➤ 5. IMPLEMENTATION

The implementation of the project involved developing a static website using **HTML, CSS, and JavaScript** and hosting it on **Amazon S3**. An **AWS account** was created, and an **S3 bucket** was configured with the **Static Website Hosting** feature enabled. The website files were uploaded to the S3 bucket, and the required bucket policies were configured to allow public access. **GitHub** was used to store and manage the project source code through version control. Finally, the website was accessed using the **Amazon S3 Website Endpoint** and tested to ensure that all pages and resources functioned correctly.

6.1 AWS Account Creation

An AWS account was created to access Amazon Web Services and its cloud resources. The account was configured with the necessary details to use Amazon S3 for website hosting. AWS Identity and Access Management (IAM) was used to securely manage user permissions and access. The AWS Management Console provided an easy interface for creating and configuring cloud resources. This account served as the foundation for implementing the project.

6.2 S3 Bucket Creation

An Amazon S3 bucket with a unique name was created to host the static website. The bucket was configured in the selected AWS region based on project requirements. Static Website Hosting was enabled to allow the bucket to host website files. The **index.html** file was set as the default homepage, and an error document was also configured. The bucket served as the storage location for all website resources.

6.3 Bucket Policy Configuration

A bucket policy was configured to provide public read access to the website files. The policy allowed users to access the hosted website through the Amazon S3 Website Endpoint. Appropriate permissions were applied while ensuring secure management of AWS resources.

Public access settings were configured according to AWS recommendations. This ensured secure and successful website hosting.

6.4 Website Upload

The website was developed using **HTML, CSS, and JavaScript**. All project files, including images and other resources, were uploaded to the Amazon S3 bucket using the AWS Management Console. The folder structure was maintained to ensure proper loading of all website resources. After uploading, the files were verified to confirm successful deployment. The website content became available through the configured S3 bucket.

6.5 Static Website Hosting

The **Static Website Hosting** feature was enabled in the Amazon S3 bucket settings. The website endpoint generated by Amazon S3 was used to access the hosted website. The **index.html** file acted as the homepage, while the error page handled invalid requests. The website was tested to verify accessibility and performance. This feature allowed users to access the website through any web browser.

6.6 GitHub Repository

A **GitHub repository** was created to store and manage the project's source code. All website files were uploaded to the repository using Git. GitHub helped maintain version history, organize project files, and simplify future updates. It also served as a secure backup of the website source code. The repository improved project management and collaboration during development.

➤ 6. RESULTS

The implementation of the **Static Website Hosting Using Amazon S3 with GitHub** project was completed successfully. The following results were achieved:

- The static website was successfully hosted using **Amazon S3**, providing a reliable cloud-based hosting solution.
- The website was accessible through the **Amazon S3 Static Website Endpoint**, allowing users to view the website over the internet.
- All website resources, including **HTML, CSS, JavaScript, images, and other static files**, were uploaded successfully and displayed without errors.
- The website demonstrated **fast page loading** and consistent performance due to Amazon S3's optimized object storage architecture.
- Amazon S3 provided a **cost-effective, scalable, and highly available** hosting platform suitable for static websites of different sizes.
- Public access permissions and bucket policies were configured correctly, enabling users to access the hosted website securely.
- **GitHub** was successfully used as the version control system to store, organize, and manage the website source code.
- The repository maintained a complete history of project updates, making it easier to track changes, collaborate, and restore previous versions when required.
- Website updates could be performed easily by modifying the source files locally, committing the changes to GitHub, and manually uploading the updated files to the Amazon S3 bucket.
- The project required **no web server or backend infrastructure**, reducing deployment complexity and maintenance efforts.
- The hosting solution proved to be highly reliable, with minimal administrative overhead and no server management responsibilities.
- The project demonstrated how cloud storage services can be effectively used to host static websites while leveraging GitHub for efficient source code management and version control.

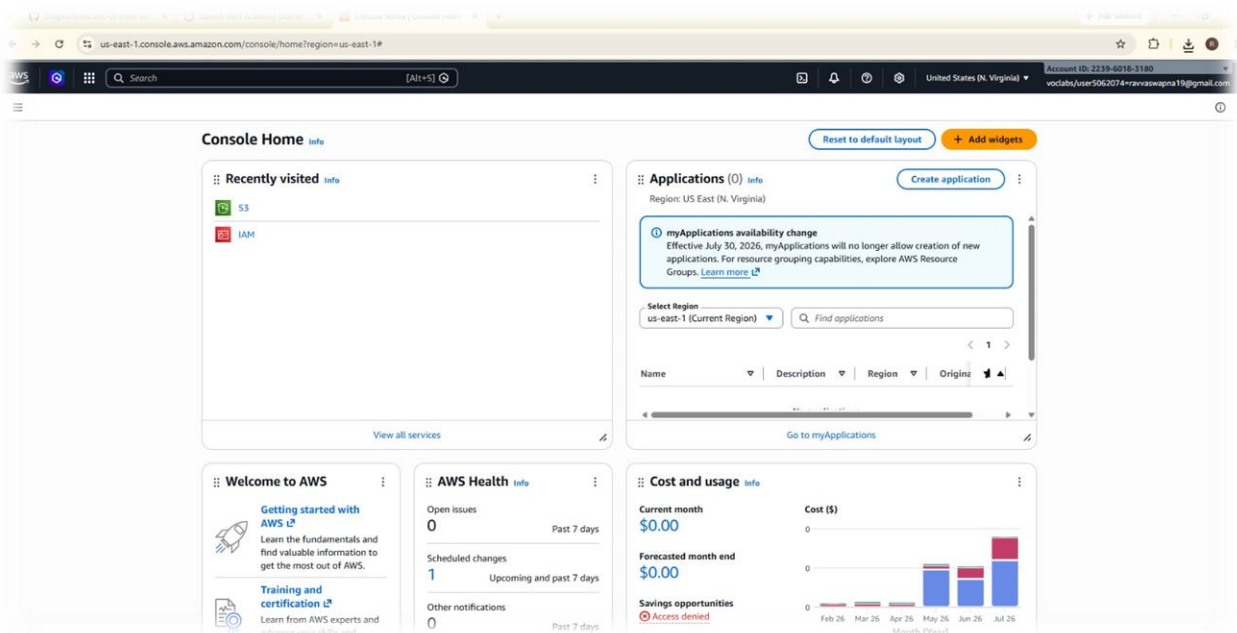
- The implementation met the project objectives by providing a secure, scalable, and easy-to-maintain static website hosting environment without the use of CI/CD automation.

Result Summary

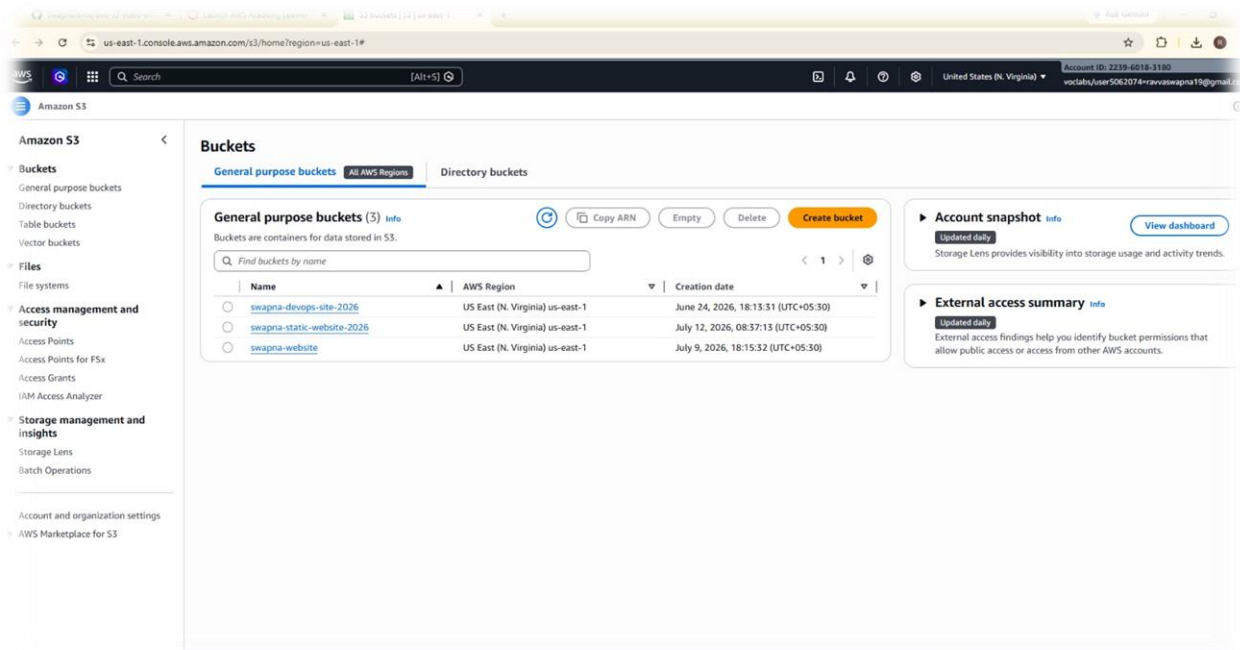
The project successfully demonstrated the deployment of a static website using **Amazon S3** and the management of website source code through **GitHub**. The website was hosted efficiently, delivered reliable performance, and was easy to update through manual uploads, making the solution simple, cost-effective, and suitable for educational, personal, and small business websites.

Screenshots of the output

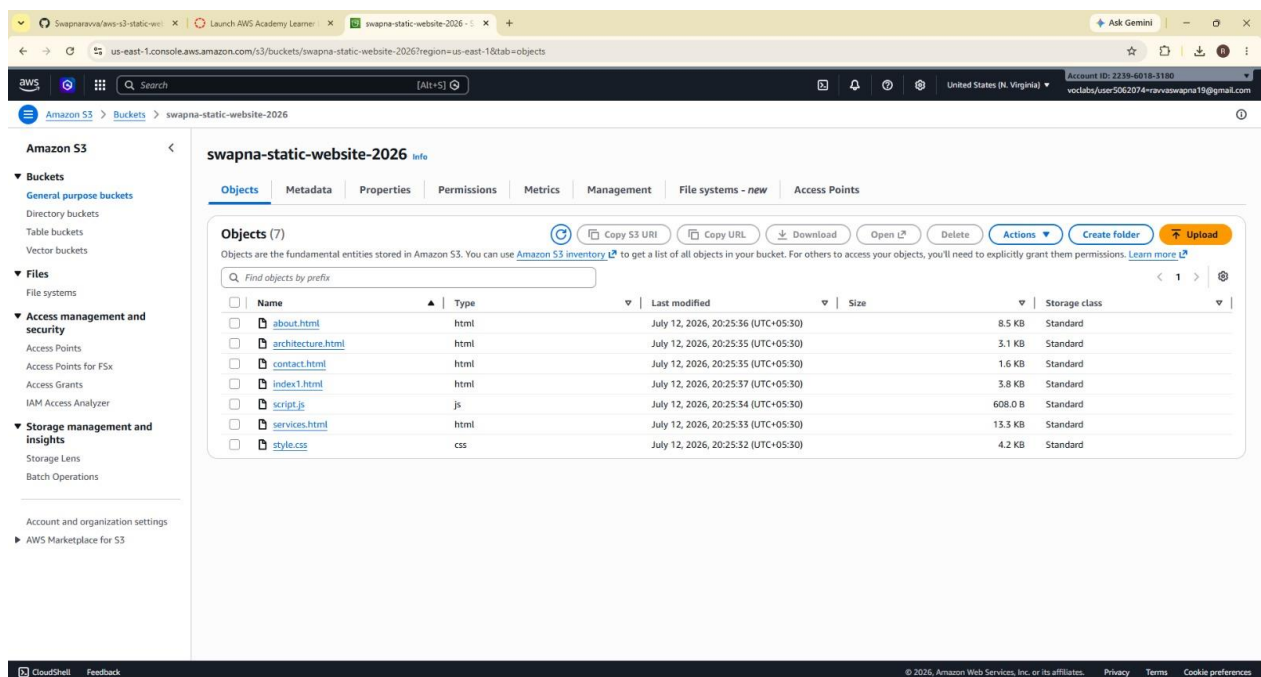
6.1 Amazon console



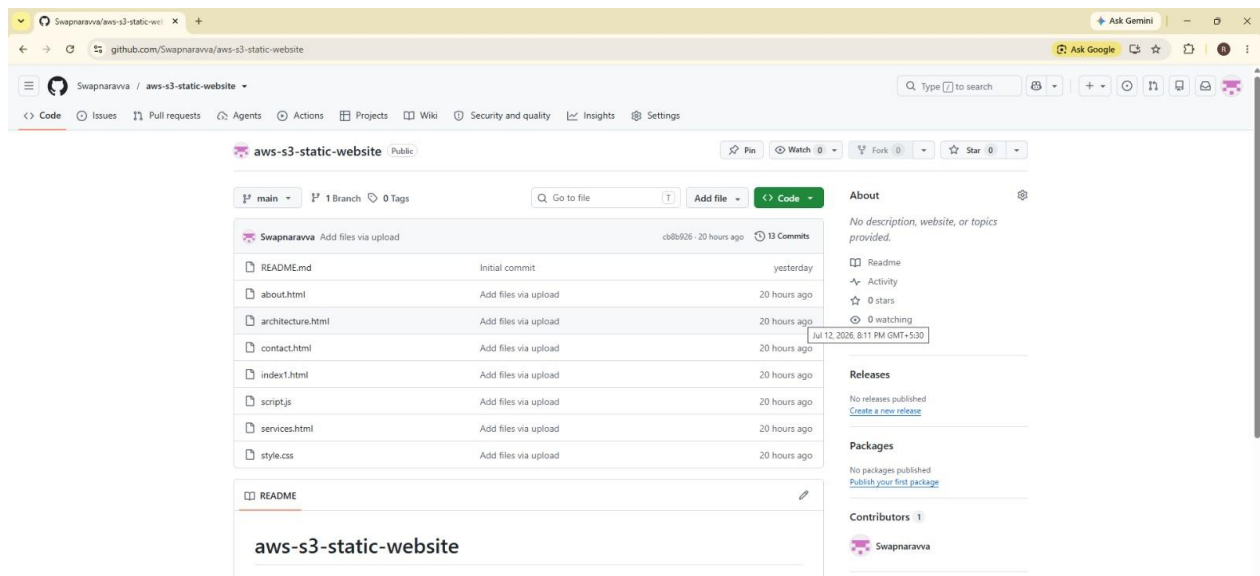
6.2 Buckets



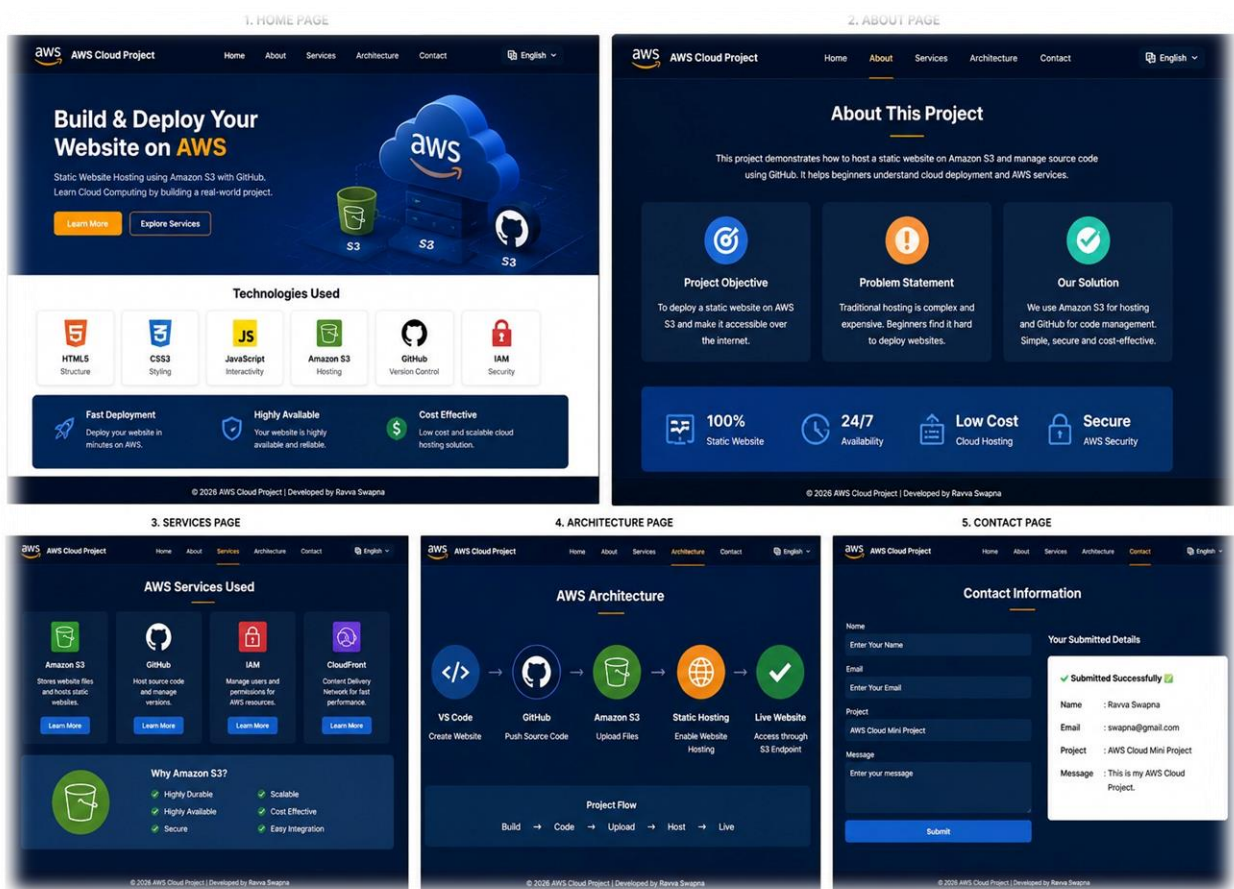
6.3 Uploaded Files into Amazon S3



6.4 Git Hub



6.5 Output Screen shots of the Static Website Hosting using Amazon S3



The above figure shows the successful implementation of the static website hosted using Amazon S3 with GitHub for source code management. The website consists of five main pages:

- **Home Page** – Displays the project overview, technologies used, and key features.
- **About Page** – Explains the project objective, problem statement, and proposed solution.
- **Services Page** – Presents the AWS services used, including Amazon S3, GitHub, IAM, and CloudFront.
- **Architecture Page** – Illustrates the workflow from website development to deployment on Amazon S3.
- **Contact Page** – Provides a contact form and displays the submitted user information.

➤ 7.CONCLUSION

The **Static Website Hosting Using Amazon S3 with GitHub** project was successfully implemented, demonstrating an efficient and reliable method for hosting static websites on the AWS cloud. Amazon S3 provided a secure, scalable, and cost-effective platform for storing and delivering static website content, while GitHub served as an effective version control system for managing the website's source code.

Throughout the project, the website was developed using **HTML, CSS, and JavaScript**, uploaded to an Amazon S3 bucket, and made publicly accessible through the S3 static website endpoint. The implementation verified that all webpages, images, stylesheets, and scripts were hosted and rendered correctly, ensuring smooth navigation and a responsive user experience.

The project also highlighted the advantages of cloud-based static website hosting, including **high availability, low maintenance, scalability, and reduced hosting costs**. By using GitHub for source code management, project files could be organized efficiently, tracked through version history, and updated whenever necessary.

Overall, the project successfully achieved its objective of hosting a static website using Amazon S3 while managing the source code with GitHub. It provides a strong foundation for understanding AWS cloud services and can be extended in the future by integrating services such as **Amazon CloudFront** for faster content delivery, **Route 53** for custom domain management, **AWS Certificate Manager (ACM)** for HTTPS support, and **GitHub Actions** for automated deployment.